



How ethical areas of post-normal science can invigorate mathematics education

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ABSTRACT

Future experts, decision-makers, stakeholders and future citizens – they all go to school, which could be a place to prepare students to facing post-normal problems in their future roles. While the literature on mathematics and science education shows increased interest in post-normal science to critically understand contemporary problems, mathematics is still generally taught as a neutral subject with either correct or wrong answers. In this paper, I lean on post-normal science literature and the slowly growing literature on ethics and mathematics education to develop a framework for identifying how mathematics education can prepare students for ethical aspects of post-normal problems. Three areas of ethics associated with post-normal problems are explored through concepts from ethics and mathematics education: ethical perspectives of the problem, ethical perspectives embedded in knowledge, and ethics in interpersonal care in discussions and care for students' self-esteem in participatory processes. This framework is used to analyse and discuss three classroom projects to provide examples of what these areas may contain. The projects partly align with post-normal science, and are about clothing and sustainability, traffic safety, and argumentation related to open an offshore area to oil exploitation. The ethical areas are discussed in terms of ethical awareness and democratic practices.

1. Introduction

The world is facing numerous complex problems that are associated with uncertain knowledge, high stakes and urgency, but with no obvious best solution. Problems with these characteristics, such as climate change, loss of species and pandemics, are often called post-normal situations (Funtowicz & Ravetz, 2003) post-normal problems or wicked problems. The understanding of such problems are often contested harshly and can result in polarisation.

Post-normal science offers a general approach to these kinds of problems: (i) placing uncertainty at the centre of the discussion on how to handle the problem, (ii) giving more attention to stakes and the relevance of supporting knowledge in quality assessments, (iii) embracing a broad range of knowledge forms, and (iv) including perspectives of stakeholders, risk bearers, and citizens in general in decision-making processes (Funtowicz & Ravetz, 1994). The latter point is referred to as an extended peer-review as it is about ensuring quality of knowledge and perspectives seen in relation to decision-making. Extended peer-reviews can be conducted in several ways (for an overview, see O'Connor & Douguet, 2024, same issue). Common for these approaches is the emphasis on inclusion and a broad involvement of stakeholders. Central to the decision processes is mapping stakeholders' decision preferences and mapping impacts of the different decision options on the various stakeholder interests. This then requires a process on discussing and deciding

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on quality measures and giving input on these (see e.g. O'Connor & Douguet, 2024; De Marchi et al., 2000; Giampietro & Bukkens, 2015).

Post-normal science thus operates in several ethical areas that are partially interdependent. Firstly, a post-normal problem is ethical in itself, as choices of handling the problem will affect people's lives and/or the environment in different ways in terms of pros and cons. The limited political response to climate change is an example, where the decision on not to act substantially favours consumers in the northern parts of the world and disfavours those who live in areas where climate change has a significant impact. A second ethical area in post-normal science is bringing attention to ethics being hidden in so-called value-neutral science methodology. Furthermore, the idea of extended peer-reviews, i.e., participatory processes towards decision-making and the inclusion of a plurality of knowledge forms, introduce another ethical area: the ethical and moral treatment of the extended peer-reviewers, respecting legitimate perspectives and the human dignity of the involved.

Everybody with a role in post-normal science has gone to school; experts, policymakers, stakeholders and citizens in general. This makes teaching and learning in school relevant for post-normal science, but also the other way around as ideas from post-normal science are relevant for education. In this paper, I explore the role ethical areas of post-normal science can have in mathematics education, in particular from a perspective of extended peer-reviews. The basis for this includes ideas and concepts from post-normal science and critical mathematics education, in particular literature on ethics and mathematics education. This results in a framework of ethics, which is used to analyse and discuss three classroom projects. The projects are on clothing and sustainability, traffic safety and argumentation related to open an offshore area to oil exploitation. The projects are chosen because of their similarities to post-normal problems.

In the following, I present components in curricula that support post-normal approaches to show how post-normal science can be considered relevant in education as schooling is understood today. Then follows education literature which connects post-normal science and the research field of critical mathematics education to show what has been achieved and to indicate how ethics is only implicitly addressed in this literature. Because ethics is the focus of this paper, a section on mathematics education and ethics is included. This serves as one of the framework pillars, while ideas from post-normal science serve as the other. Then the framework is presented, which purpose is to identify ethical aspects of students working with post-normal problems in mathematics education. To provide further explanation of the framework, it is illustrated through the three school projects. Finally, key points are summarised and discussed more broadly.

2. Mathematics education and post-normal science

The purpose of this section is to show that including ideas of post-normal science can be considered relevant in school according to some curricula and to show how mathematics education and post-normal science is connected in research.

2.1. Post-normal science and curricula

The Norway curriculum includes several elements that to some extent are in line with post-normal science. As a general value, democratic practices is highlighted, both in society and in classrooms (Ministry of Education and Research, 2017). It states, "A democratic society is based on the idea that all citizens have equal rights and opportunities to participate in the decision-making processes." This implies that education has a role of giving the students experiences that resemble extended peer-reviews. In addition, the curriculum puts an emphasis on understanding that controversial contemporary societal and environmental problems may not have simple or single solutions. This means that the students should learn that disagreement is legitimate when problems are complex, such as post-normal problems. Furthermore, it says,

"The school must be a venue where children and young people experience democracy in practice. The pupils must experience that they are heard in the day-to-day affairs in school, that they have genuine influence and that they can have impact on matters that concern them. They must gain experience and practise different forms of democratic participation in the day-to-day work with their subjects" (Ministry of Education and Research, 2017).

This is significant for students' experience in having a say and for achieving the confidence to participate in extended peer-review communities. In addition, *democracy and citizenship* is a multi-disciplinary topic that concerns all school subjects (Ministry of Education and Research, 2017), which is another reason for including post-normal problems in teaching and learning.

The Organisation for Economic Co-operation and Development (OECD) addresses similar values and goals to some extent (OECD, 2019). For example, critical thinking, responsible citizenship, social capacities and learning from the young are emphasised topics but are more related to individual needs and workforce needs, and less to critical citizenship compared to the Norway curriculum. However, OECD also states that,

"In a world of interdependency, finding solutions to global challenges requires the ability to handle tensions, dilemmas and trade-offs – for instance, between equity and freedom; autonomy and solidarity; efficiency and democratic processes; ecology and simplistic economic models; diversity and universality; and innovation and continuity. This requires the skill of balancing seemingly contradictory or incompatible demands." (OECD, 2019 p. 63).

This means that post-normal problems and post-normal science can be regarded as relevant for OECD as well.

The mathematics curriculum of Norway (The Norwegian Directorate for Education and Training, 2019a) includes at least three statements about what students should be able to do at various levels and which aligns with post-normal science: "interpret and

critically evaluate statistical representations found in media and the local community”; “explore and argue for how representations of numbers and data can be used to promote different points of view”; and “The subject shall make the pupils conscious of the underlying conditions and premises for the mathematical models which form the basis for decisions in their own lives and in society.” Taken together, this supports the idea of critical citizenship and preparing students for extended peer-reviews. In particular the latter two statements also support the idea that quantified information may be value-laden. Related statements are found in the natural science curriculum ([The Norwegian Directorate for Education and Training, 2019b](#)), for example “Knowledge about the interaction between nature, individuals, technology and society can promote the pupils’ ability to think critically and help them make conscious decisions in their daily lives.” Uncertainty in these documents is explicitly treated through learning aims in statistics, and implicitly treated in the showed statements. Norway has long traditions of understanding education as an important arena for democratic formation, as of building and sustaining Norway as a democratic society ([Slagstad et al., 2011](#)).

2.2. Post-normal science in classrooms and in research

In spite of the intentions of the presented curricula, research suggests that democracy and citizenship receives limited attention in mathematics education as, for example, a study on teachers’ feedback to students in mathematics classrooms in Norway are mainly on students’ mathematical procedure skills rather than on engaging the pupils in mathematical practices, such as applying mathematics on societal issues ([Stovner & Klette, 2022](#); [Stovner et al., 2021](#)). While the Norway curriculum looks promising on paper, less is happening in practice other than exceptional examples. The same seems to be the case in other countries. Working with complex and controversial issues in mathematics classrooms is rare.

In educational research, on the other hand, post-normal science receives increased attention, in particular within science education and education for sustainable development, but also to some extent within mathematics education. A denominator of educational research related to post-normal science is, of course, an emphasis on the characteristics of post-normal science and how knowledge production cannot be seen as a value-free enterprise in post-normal problems. This has resulted in a call for working with post-normal problems in school to engage students in contemporary problems and prepare them for their future lives (see e.g., [Barwell, 2013, 2018](#); [Block et al., 2018, 2019](#); [Barwell & Hauge, 2021](#)).

Similarly, education literature has addressed how mathematics and mathematics education have the appearance of falsely being value-free. For example, [Skovsmose \(1994\)](#) refers to *the formatting power of mathematics* when discussing how mathematics contributes to shape the world and our understanding of it, for example through applying mathematics to develop technology and various forms of distributions of goods in society. Relatedly, [Skovsmose \(2020\)](#) applies the term *performative interpretations of mathematics* when describing how mathematics has the power to form worldviews and actions. The choice of what mathematics education should include and how the aura of mathematics is presented is not value-free either. For example, [Atweh and Brady \(2009\)](#) argue that mathematics education can have three societal roles: *conforming the society* through preparing students for their future jobs; *reforming the society* through preparing students to engage critically in the society; and *transforming the society* through providing students’ transforming capacity. National curricula and teachers, likewise, put different emphasis on the three roles and on value-laden aspects of mathematics, implying that mathematics education is not value-free. Because the last two roles of mathematics education are associated with change, critical thinking and creativity, they include the potential to educate students to participate in extended peer communities, in line with post-normal science.

Mathematics is an essential tool for science, in particular for making science sound and transparent. Assessing uncertainty and soundness of quantified information and considering the role of these are an important part of post-normal science ([Funtowicz & Ravetz, 1990](#)). Furthermore, mathematics is also used for other purposes in society than as a tool in science, which involve substantial uncertainty, high stakes and urgency. Examples include GDP (gross domestic product), distributions of goods and taxes and a range of indicators. These are based on mathematical models which inform decision-making, so that these situations are similar to post-normal problems. This means that ideas from post-normal science are relevant also for policymaking where science is not an ingredient.

Post-normal science has received limited attention in mathematics education, in spite of the importance of mathematics in post-normal problems. Here follows a brief summary of what aspects of post-normal science have been highlighted and how. What the papers have in common is first of all a description of post-normal situations or problems: complex and urgent socio-political problems which require science-based advice, but where stakes and interests are conflicting and “facts” are uncertain (see for example [Barwell, 2013](#); [Barwell & Hauge, 2021](#); [Hauge, 2022](#); [Hauge & Barwell, 2017, 2022](#); [Steffensen, 2017](#)). Climate change and other problems within sustainable development are highlighted in these texts to illustrate the relevance of post-normal science. Including post-normal science and associated problems into mathematics classrooms breaks with traditional mathematics education where the students’ answer is either correct or incorrect because there is only one single correct answer.

As pointed out above and elsewhere ([Barwell, 2013, 2018](#); [Hauge & Barwell, 2017](#); [Steffensen, 2017](#)), critical mathematics education and post-normal science share some similar ideas, in particular the need to be aware of how choices of data and methods may impact users or stakeholders in hidden ways. This implies that values may be embedded in knowledge production and needs to be addressed at various levels: defining the problem, the knowledge production and decision making. The solution to this problem in post-normal science is extended peer-communities, suggesting a more democratic approach to decision-making where stakeholders can have a say. This aligns with critical mathematics education with emphasis on preparing students for critical democratic citizenship through engaging them in mathematics-based argumentation in discussions about socio-political topics ([Barwell, 2013, 2018](#); [Hauge & Barwell, 2017](#)). Mathematics can then be a tool to investigate injustice, ecological conditions etc., or the use and role of mathematics can be scrutinised in terms of how uncertainty is handled.

A couple of papers on mathematics education and post-normal science include ideas on what pedagogical methods can be useful for

preparing students for extended peer-reviews. Key qualities that have been highlighted include working with authentic problems involving enquiry-based learning and where democratic practices are facilitated in the classroom (Barwell & Hauge, 2021; Hauge & Barwell, 2022). These features are important as a preparation for joining extended peer-communities. Such teaching and learning situations could actually represent extended peer-reviews if the students represent multiple views and their contribution is included in decision-making. However, such situations may be quite different from being exposed to the peer-review frameworks described by O'Connor and Douguet (2024, this issue).

The handling of uncertainty is an important perspective in post-normal science as how it is dealt with may affect stakeholders in different ways. Because statistical measures are often not sufficient for representing uncertainty, Funtowicz and Ravetz (1990) developed three categories of uncertainties: (1) Inexactness, where statistical measures are sufficient for assessing uncertainty; (2) Unreliability, where simplifications and other choices in knowledge production can question the sufficiency of statistical measures; and (3) Border with ignorance, where there may be known and unknown unknowns. Only a few papers on mathematics education have included this typology of uncertainties. Concerning questionable numbers in the media, the categories have been considered useful to point out that invalid quantified information is not necessarily disinformation by intention, but may just be uncertain (Hauge, 2022). The typology can thus increase an awareness of non-quantifiable uncertainty. The categories have also been used to analyse students' awareness and understanding of non-quantifiable uncertainty (Hauge & Barwell, 2017; Hauge & Herheim, 2015).

This paper addresses mathematics education, but Post-normal science has been addressed in similar ways in science education and education for sustainable development. Key topics have been risk society, uncertainty in science for policy, citizen involvement and what this requires of education and curricula (see e.g. Aikenhead, 2022; Colucci-Gray & Camino, 2016; Wals, 2012).

Ethics has not been part of the literature on post-normal science and mathematics education, only implicit in that post-normal science have ethical implications and consequences. In general, there is limited research on ethics and mathematics education, in particular related to socio-scientific problems. In order to connect ethics to post-normal science in education, I now present an overview of literature on ethics and mathematics education that is relevant for connecting post-normal science and mathematics education.

3. Mathematics education and ethics

Mathematics education is ethics-based as its purpose is to benefit both the individual learner and the society (Atweh et al., 2012). Learners need mathematical skills in their everyday life, in their future jobs and as democratic citizens. Further, mathematics education is concerned with *good practice*, *good achievement* and *good assessments*, but the ethical *good* behind these topics are less discussed (Atweh, 2013). For example, should good practice be connected to advancing future engineers or to fostering democratic citizenship?

Similar to in the academic field of post-normal science, scholars within critical mathematics education have given attention to how values can be embedded in mathematics-based decision-making. This shows that mathematics applied in society has an ethical dimension. Yet only a small number of academic publications connects such aspects explicitly to ethics or to theory of ethics, and this limited literature is from the present millennium (Dubbs, 2020). Several of these works lean on Emmanuel Levinas' concept of relation with the *Other* (e.g., Atweh & Brady, 2009; Atweh et al., 2012; Ernest, 2012), addressing the teachers' responsibility of developing constructive relationships between teacher and students, and addressing how to facilitate students' responsibility.

Atweh et al. (2012) discuss ethical implications of the premise that mathematics education has a role in developing students' social responsibility and preparing them for democratic citizenship. They criticise critical mathematics education for concentrating on criticism of injustice and not giving attention to solutions, as students need an awareness of what and how things can be changed in relation to the *Other*. In other words, attention is given on what justice is (how things ought to be) and on knowledge on injustice (what people ought to know about things), and little on solutions (ought to do about things). They lean on Levinas when they argue that although ethical responsibility towards the *Other* is fundamental, it is not possible without knowledge and politics. This means, they continue, that the *ought to do* is based on the *ought to know* which in turn is based on the *ought to be*. Giving attention to what students and what society ought to do can move focus from criticism to visions and a commitment to action. This action could be related to extended peer-reviews, where *ought to do* could be about suggesting solutions to post-normal problems; *ought to know* could be insights of the problem, politics and knowledge-production; and *ought to be* could be related to facilitating students development of capacities and virtues they benefit from when joining extended peer-review communities.

Levinas' concept of relation with the *Other* has also been used about encounters in the classroom. For example, Boylan (2016) argue that while addressing socio-political topics in a mathematics classroom may prepare students for democratic engagement, such topics can also favour students from intellectual homes (Boylan, 2016). Because of complexities, ambiguity and dilemmas, considering an act or an encounter as ethical or not often depends on the situation. To reflect on this complexity Boylan (2016) highlights that there may be different and competing ethical commitments in the classroom. He presents four ethical dimensions of mathematics education to conceptualise this. The first dimension is *being with others*, which denotes a responsibility for recognising that students are different and that to develop their sense of purpose may necessitate different approaches and activities. This requires a dialogical relationship. The second dimension is the *societal and cultural dimension*, which implies a responsibility towards social justice and democratic engagement. The third dimension, the ecological dimension, is on recognising how mathematics has contributed to current environmental crises, but also its role in responding to them. The last dimension is related to the student and the teacher's self-appreciation: the *Self*. Dubbs (2020) argues that Boylan's dimensions can be seen as an extension of Levinas' concept of relation with the *Other* where the environment and society can be understood as the *Other*. It can also be argued that the dimension on *Self* is an extension as well, as it can be understood as an engagement with the *Other* in encouraging their *Self*. Boylan continues that mathematics education has the responsibility to encourage students to develop responsibility in each of the three first dimensions, including facilitating students'

reflexivity related to them.

Looking at these dimensions from a post-normal science view, the dimensions related to the environment and society are relevant for showing care and responsibility to post-normal problems together with its possible ethical consequences. Because the dimension on the *Other* and the *Self* are more focused on individuals and encounters with individuals, the dimensions are more relevant for processes in extended-peer communities. The *Other* can also represent risk-bearers of post-normal science.

Other scholars have discussed similar ethical perspectives as Atweh et al. (2012) and Boylan (2016). For example, Ernest (2019) points to possible harms due to mathematics and mathematics education, where traumatising students with demands on mathematics skills constitutes one of the harms. Similar to post-normal science literature, he also refers to harm caused by certain applications of mathematics in society, such as in developing destructive technology and applications of IQ-tests. Together with colleagues, I (Hauge et al., in press) show how including controversial socio-political issues in teaching and learning, such as post-normal problems, can be sensitive to some students and thus requires particular care for these students as the *Other* and concern for their, and their teacher's, *Self*. Extended peer-review communities also benefit from showing care and respect for each of the contributors and their contributions to ensure quality through a democratic process. As well as in mathematics education dealing with post-normal problems, extended peer-reviews may deal with ambiguity and tensions between the dimensions, which calls for flexibility. The ethical dimensions are intentioned to support an ethical praxis through navigating between the dimensions (Boylan, 2016).

4. Ethical areas

Based on the above discussions, I here present a framework of ethical areas that are relevant for teaching and learning when working with post-normal problems. The first area relates to the post-normal problem itself through exploring ethical perspectives of the stakes. Depending on the decision, it has impact on people and/or the environment. Decisions may result in damage on nature or something else that is valued, and people may find decisions unfair. The first area is thus an ethical awareness of such impacts, which requires to identify stakes, stakeholders, risk-bearers, risks and other consequences of decisions. Boylan's (2016) three dimensions related to society and culture, environment and *Other* are all relevant, as these serve as representing stakeholders and what is at stake concerning possible impacts.

The second area concerns how values may be embedded in the knowledge production; in the questions asked, in the methodology and how results are interpreted. This is closely related to understanding how assumptions and uncertainty have ethical consequences, as choices on how these are handled may produce results that favour and disfavour certain stakes and stakeholders. Skovsmose (1994) has presented similar thoughts through his notion of *the formatting power of mathematics*, the power quantified information and distributions have on the world and how the world is perceived. For example, it has been pointed out how the Individual Biomass Index (BMI) was developed for health purposes, while it is very often used individually as a measure for slimness, in spite of the considerable associated uncertainty. Teachers who consider working with the index in mathematics class, point to this topic as an ethical challenge, but also to sensitivity of bringing it up in the classroom (Hauge et al., in press). Boylan's (2016) dimensions related to society and culture, environment and *Other* are all relevant for the second area, as they describe who and what can be affected by the embedded values, similarly to the first area. Because knowledge production in post-normal science could be by experts or locals, ethical awareness of embedded values concerns science, expert knowledge and local knowledge, including the students' local knowledge.

The third area is related to ethical perspectives in the classroom when the post-normal problem and possible solutions are explored and discussed. Key aspects are respecting fellow perspectives and fostering democratic virtues. This activity resembles, or the activity could actually be, an extended peer-review. Ethical concerns in such classroom activities are therefore similar to desired ethical concerns in extended peer-review communities. Boylan's (2016) ethical dimension *Other* is relevant, but while *Other* in the first and second area relates to stakeholders and risk-bearers, *Other* relates to the members of the extended peer-review community or the classroom. In addition, Boylan's (2016) dimension *Self* is significant. Post-normal science depends on citizens with confidence to participate in discussions, so that facilitating students' self-confidence and self-appreciation in similar situations is important. .

5. Methods

Empirical data from three research projects have been chosen to show how various aspects of the ethical dimensions can play out in practice. None of the projects were planned to be related to ethics or post-normal science. Data from all projects are transcribed, anonymised, collected and handled in accordance with Norwegian and institutional ethical requirements and standards. This implies signed consent forms by the research subjects and the students' parents, encrypted storage of recordings, and recordings were deleted

Table 1

The three ethical areas of post-normal science and how they connect with Boylan's (2016) ethical dimensions. The shaded areas indicate key ethical relations involved in each area.

Ethical areas	Nature	Society	Other	Self
Stakes in post-normal problems				
Impact of values embedded in knowledge				
Interpersonal care in extended peer reviews				

after being transcribed. The students who had not signed consent forms were present in the classrooms but were not recorded.

The first project was on four preservice teachers in their practicum who were given the task by researchers (including me) and their practicum teacher to plan and carry out a three-week school project on sustainable development in all mathematics, natural science and social science class. The preservice teachers chose clothing and textiles as the sustainability topic. There were twenty-three students in the class, ninth grade, 14–15 years old. Data include video and sound recordings of lessons and student assignments. This project is included in this paper as the issue of clothing and the textile industry is a post-normal problem and because the complexity of this problem is illustrated in the project together with the injustice of involved actors. In addition, the students are given the opportunity to have a say, and there are ethical implications of the project.

In the second research project, a mathematics teacher of eighth grade invited researchers to observe and study her project on risks associated to traffic close to the students' school. The project lasted a couple of weeks, but data consist of video and sound recordings of only the last day of the project, when the students work in pairs with how to present the data, and where the class reflected on the project in the very final stage. The teacher was interviewed to get information on the previous parts of the school projects, for example how the students collected data and the role of students in developing the project. More information on the project can be found elsewhere (e.g., Hauge & Herheim, 2015; Herheim & Rangnes, 2015). This school project is included because it shows an example of how students are involved in thinking about risk and ethical consequences, and because the project has an impact on the local community, which can resemble a process of an extended peer-review.

The third research project involved around 50 students in Lofoten, eighth grade, discussing whether to open the areas off the Lofoten islands to petroleum exploitation. The school project lasted one day and was carried out by a research team, which collaborated with one of the teachers in developing the school project. Data consists of video and sound recordings of group discussions and the plenary discussion. More information on the projects can be found in several chapters in the books edited by Breivega and Rangnes (2019) and by Herheim, Werler, & Hauge (2022). This discussion is included as the issue is a post-normal problem and because the students managed to reveal characteristics of post-normal problems and that different decisions would impact stakes differently. In addition, although there was disagreement, the students showed respect for each other and others' opinions.

6. Ethical areas in practice

I now turn to three examples of students working with post-normal problems in classrooms to explore how the ethics areas can be applied in practice.

6.1. Student project on sustainable development

Clothing and textiles can be argued to be a post-normal problem as it represents an urgent, considerable and complex environmental and social problem with conflicting stakes and uncertain facts. Four preservice teachers conduct a three-week project on clothing and sustainable development during their practicum where they introduce some of this complexity. Among other things, they point to several ethical problems. Through lessons on textiles, poverty and working conditions in China, the students learn that textiles cause pollution, and that the clothing industry can be harsh, both because of low salaries and because of harmful chemicals. The students also watch a TV programme that underlines that because of such conditions, people in Norway can buy cheap and polluting clothes. Although the teaching and learning situations do not specifically refer to stakes, the students do get an impression of a problem where there are risk-bearers and where both the industry and consumers gain from the situation today. This implies that the students are involved in ethical considerations within Area 1 (see above), where they get an impression of several risk bearers: Societies suffer from unfairness, individual workers may suffer from poor working conditions and the environment becomes polluted from micro-plastics, thus three of Boylan's (2016) ethical dimensions are involved: the society, the *Other* and the ecological dimension. The students also get an impression of competing stakes.

After some introductory lessons, the students get to choose their own project within clothing and textile industry. They were asked to also find information on the internet. However, the preservice teachers noticed that the students were uncritical to quantified information they found on various web sites, so the preservice teachers dedicated two mathematics lessons to critical thinking related to graphs in the media. Thereby the students became acquainted with uncertainty to some extent, but the link between uncertainty and ethical aspects connected with stakes were not made explicit. This suggests that if students reflected on ethical consequences of handling uncertainty in knowledge production, that is, within Area 2, this happened on their own.

Concerning Area 3, on showing consideration for the students in the classroom situations as the *Other* and care for their *Self*, giving the students the possibility to choose their project, the preservice teachers recognise possible interests the students might have. For example, one group of students chose fast fashion because they were concerned with the problems it caused. Through their project, they were given the opportunity to present the problems with fast fashion to the whole class. The preservice teachers and the fellow students showed care for the group as the *Other* since the floor was given to the group, and their fellow students seemed to listen carefully. This way, the classroom environment supported the group members' self-appreciation regarding their views. In addition, one of the group members participated in a role-play debate in the classroom on fast fashion, where she was given the opportunity to present argumentation against fast fashion.

However, when the students' projects were graded, little emphasis was placed on value perspectives and ethical considerations that were implicit in the projects. Rather, the facts, the weighting between the school subjects, answering the questions and presentation techniques were addressed. For example, the only feedback with an ethical touch given to the group who had worked with fast fashion, was that it was nice that the group had addressed problems with fast fashion. So, while the projects gave the students an opportunity to

voice their points and views, the grading of the projects seemed to be counter-productive in appreciating the students' engagement. It is perhaps not surprising that value-perspectives and students' formation receive less attention in grading, but grading can thereby contribute to harm students' self-appreciation in their role as extended peer-reviewers despite activities that aim to facilitate their self-appreciation (see also [Hauge et al., in press](#)).

6.2. Traffic project

A mathematics teacher in lower secondary school finds the number of cars that have driven off the road and into the sea alarming. This is a stretch where the school bus passes every day, and a car driver had drowned the previous year. The teacher develops a project together with her students where they measure the road barriers and do a traffic count. The teacher provides a table with vehicle categories which the students can fill in during their traffic count. The students are invited to discuss and decide how to measure the barriers in terms of the distance between each measurement and how to measure the height, since the angle of the barriers is not 90 degrees. With assistance from the police, the students and the teacher collect their data. The local newspaper reports the event, and one of the students is interviewed. Back in the classroom, the students work with the data and how to present them in an adequate manner. They find that the height of the barriers is lower than Norwegian traffic regulations. When working with presenting the traffic count, two students working together oppose to what is recorded in the category Other vehicles. They argue that wild sheep in the road should have its own category as they signify risk. Another student argues that the category Large vehicles should be divided into two categories – bus and lorry – because if a bus drives off into the sea the disaster potential is much greater. A month or two after the collecting of data, the road barriers are replaced with new and higher ones. However, the teacher does not know whether their project caused this change.

This example is perhaps not a post-normal problem, as the problem is not complex and the stakes not conflicting, although people can die and cars get damaged, and improving the barriers require resources. However, the example illustrates ethical perspectives that are different from the previous example, and which falls within Area 3. The problem itself is ethical as people have died, vehicles have been damaged and more people can die. The teacher invites the students to find ways to express the risks. The students contribute by deciding the distance between each measurement of the height of the barriers, they collect the data, and they are allowed to influence the way their results are communicated in the classroom. Thereby, the teacher recognises her students as the *Other* in contributing with valuable input. So does the local police, which finds the project worthwhile to support and, likewise, the local newspaper and perhaps even the local traffic authority. The students and their teacher's effort is recognised as a self-generated extended peer-review, where the students contributed to defining the problem, providing knowledge and suggesting a solution. The project was shown respect, and the students' (and the teacher's) self-appreciation was supported.

6.3. The Lofoten project

A group of researchers went to Lofoten to listen to students aged 13–14 discussing whether to open the Lofoten offshore area in Norway to oil exploitation ([Breivega, Hauge, & Tjomsland, 2022](#)), a topic which had triggered heated political debates for decades. The task given to the students was first to work in groups and find arguments related to opening or closing the area, with a suggestion that fisheries, tourism, local economy, workplaces and cultural aspects could be relevant points to discuss. The students were then engaged in a plenary discussion.

Through the discussion, the students implicitly addressed various stakes, stakeholders and risk-bearers, such as depopulation, work opportunities, consequences of oil blowouts, and the scenery in case of industry development. The students saw themselves as stakeholders and risk-bearers as they pictured a Lofoten they feared or aspired, although there was considerable disagreement among the students of what future they preferred and what opening the area to oil exploitation would bring with it. The discussion reflected that the issue was a post-normal problem and that different decisions would affect stakes and stakeholder differently, for example that deciding to open Lofoten for exploitation could harm fish, fisheries and tourism but could provide more workplaces and develop their town in other ways. It thus showed that decisions had ethical implications in terms of possible harms to nature and society and to quality of life, thus within Area 1, covering three of [Boylan's \(2016\)](#) ethical dimensions: ecological, societal/cultural and *Other*.

The plenary discussion resembled an extended peer review in some ways. Although the students were not asked to review knowledge that could be included in a decision-making process, they contributed, internally, to understand and define decision problems by bringing values and perspectives to the table. It has elsewhere been concluded that the students addressed key aspects of risk as they managed to display a complexity through including a variety of relevant topics, a range of uncertainties were revealed through expressed doubts and counter argumentation, and the students portrayed values, stakes and risk-bearers ([Breivega et al., 2022](#)). Worth noting, the discussion had qualities that are essential for extended peer-reviews. A plurality of experiences, meanings and perspectives were allowed, where the students continuously built on each other's arguments ([Breivega et al., 2022](#)). They allowed each other to disagree on values and opinions. The students thus showed, and were shown, respect, so that care for the *Other* and the students' self-appreciation, two of [Boylan's \(2016\)](#) ethical dimensions, were present throughout the discussion, implying Area 3 in the ethics framework.

7. Summary and discussion

To shed light on how mathematics education can prepare students to facing ethical aspects of post-normal problems, I developed a framework to suggest areas of what ethical considerations that might be. The dimensions are a further development of [Boylan's \(2016\)](#)

ethical dimensions considering possible recipients of care in mathematics education: *nature, society, Other, Self*. These dimensions are adapted to education for post-normal science, adding three ethical aspects of post-normal science: being aware of stakes and stakeholders of post-normal problems; considering value-ladenness hidden in the knowledge production; and the ethical encounter of the *Other* and students' self-appreciation in extended peer-reviews or processes with similarities to extended peer-reviews. The framework adds to suggested principles of education for post-normal science, suggested previously (Barwell & Hauge, 2021; Hauge & Barwell, 2022) in that ethical considerations are more explicit, in particular related to students in the classroom situation.

To illustrate and try out the ideas of the framework, I have presented three teaching and learning situations where teachers, preservice teachers and students expressed or showed various ethical concerns. What all the situations have in common is that they involve a problem that has ethical consequences and which the students addressed. In the project on textiles, ecological, societal and individual stakes were pointed to by the preservice teachers concerning the production and consumption of textiles, and in the traffic project, both teachers and students brought up societal and individual stakes. Similarly, ecological, societal and individual stakes were included in the discussion about oil exploitation (see Table 2). Only the project on textiles addressed implications of knowledge production, but this was only loosely addressed. However, the students in the traffic project experienced how choices in collecting data and choices in how the data was presented could have ethical consequences. Furthermore, the students discussing oil exploitation experienced uncertainty in perspectives and future scenarios through listening to each other's arguments. The students' perspectives and initiatives were valued in all projects, but more intentionally so in the traffic project and the oil exploitation project, as they were given opportunities to influence the course of action in the classroom. Table 2 shows which various ethical considerations were addressed, where the brackets indicate lesser focus.

The teaching and learning situations had some features in common with post-normal science. Two of the projects are on post-normal problems: textiles and oil exploitation, as they both involve uncertainty, complexity, high and conflicting stakes, urgency, and decisions lean on expert knowledge. The third project, the traffic project, is not complex, but involves risk. The projects resemble extended peer-reviews in several ways. The students are not much involved in evaluating expert knowledge, but they bring own experiences, local knowledge and own perspectives to the school projects. Because their perspectives are supported or argued against, they experience uncertainty in knowing. The textile project and the oil exploitation project include debates reflecting disagreements on problems and solutions, which is expected in extended peer reviews. The projects are not real in the sense that they result in decisions. For example, participating in a role-play debate in the textile project or presenting a project together with argumentation on the problems with fast fashion, is not the same as contributing to an extended peer-review community. However, the problems are very real, and the traffic project did result in new barriers. The students also experience ideal situations in that they show respect for each other's views. Taken together, the projects have similarities with extended peer-reviews, but in slightly different ways. The projects thus prepare students for critical citizenship and for joining future extended peer communities. Furthermore, the projects fulfil the principles of education for post-normal science suggested by Barwell and me (Barwell & Hauge, 2021; Hauge & Barwell, 2022) in that the students explore meaningful situations of risk and uncertainty, they explore scientific/mathematical concepts and societal perspectives, and they learn through dialogue.

Teaching and learning situations like the ones presented in this paper can be valuable in several ways. First of all, they support the development of critical citizenship, showing ways to contribute in extended peer-reviews or how to engage critically in society. Research shows that students who are frequently engaged in discussing controversial issues become more interested in politics and are more willing to discuss with people with other opinions (Hess & McAvoy, 2015). Second, the students learn about contemporary issues important for their lives. Third, the learning takes place in an environment that practices democratic virtues, so that the projects also serve as formation projects. Thus, they support democratic virtues, such as respect and considering others' argumentation, which contribute to making extended peer-review communities productive. Fourth, student perspectives are valued, thus, facilitating student agency, empowerment, and a sense of purpose. Empowerment in mathematics includes being able to apply mathematics or interpret mathematical representations in new real-world situations (Ernest, 2002), which is crucial for extended peer-reviews. Fifth, nurturing students' empowerment and addressing how decisions can affect stakes and stakeholders differently can enrich students' ethical awareness and ethical learning. Sixth, such projects are in line with Norway's curriculum (Ministry of Education and Research, 2017) and curriculum suggestions by OECD (2019), which highlights working with real life problems, real data and fostering democratic virtues.

Although the projects fulfil principles of education for post-normal science, outlined above, future projects can take a step further to cover even more aspects that are relevant for critical citizenship and ethical considerations. For example, the ethics can be made more explicit to the students, and students could get experience on reaching a consensus when they disagree. While revealing disagreement, understanding why and understanding how decisions affect stakeholders is key in post-normal science and education for post-normal science, the process of reaching a decision is crucial. A willingness to reach consensus in deliberative communication has been pointed out as an important experience for developing democratic citizenship by Englund (2006). In education, this could be taking local

Table 2

Areas of ethical considerations in the three teaching and learning situations, where A is the textile project, B: the traffic project, and C: the discussion on oil exploitation. Brackets indicate that the area was not significantly present in the project.

Ethical areas	Nature	Society	Other	Self
Stakes in post-normal problems	A, C	A, B, C	A, B	
Impact of values embedded in knowledge	(A, C)	B (A, C)	B (A, C)	
Interpersonal care in extended peer reviews			B, C (A)	B, C (A)

action, organising an event or writing a letter to local authorities on a neighbourhood issue. However, this does not mean that the opinions of the minority should be oppressed, but that the conflicting values and uncertainties are shown in accordance with the spirit of post-normal science.

A recommendation for future research would be to study or develop student projects where all the dimensions are prominent to see student learning outcomes, but also whether such experiences influence their empowerment and later citizenship. The presented projects did not dive deep into post-normal problems since complexities, uncertainty, quality and stakes were not explicitly addressed during the school projects. It would be interesting to see projects where understandings of these aspects are developed in depth.

Code availability

Not applicable.

Ethics approval

Yes, the projects involved are approved by ethics committee.

Consent to participate

Yes, all informants have consented to participate as research subjects.

Consent for publication

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Kjellrun Hiis Hauge: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Validation, Writing – original draft, Writing – review & editing.

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